

**TEMASEK POLYTECHNIC
SCHOOL OF INFORMATICS & IT
DIPLOMA IN CYBERSECURITY & DIGITAL FORENSICS**

AY 2026/2027 April Semester

**Project
Research Assignment**

**SERVERS ADMINISTRATION & SECURITY (CCD2C06)
SUBJECT LEVEL: 2**

GENERAL INSTRUCTIONS

1. This Research Assignment / Project consists of **6** pages (including cover page).
2. Please complete **ALL** tasks in this assignment by Week 16
3. This Project carries **40%** of the whole subject.

Content

Assessment scores	3
Group Setup and Configuration (20%) [Self-Directed Learning]	3
Group Presentation (20%)	5
Deliverables	5
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1. Assessment scores

This project contributes **40%** for the **SADS** subject. Following are the assessment components:

Assessments	Marks awarded for individual work
Group Setup and Configuration	20%
Group Presentation	20%
Total	40%

Group Setup and Configuration (20%) [Self-Directed Learning]

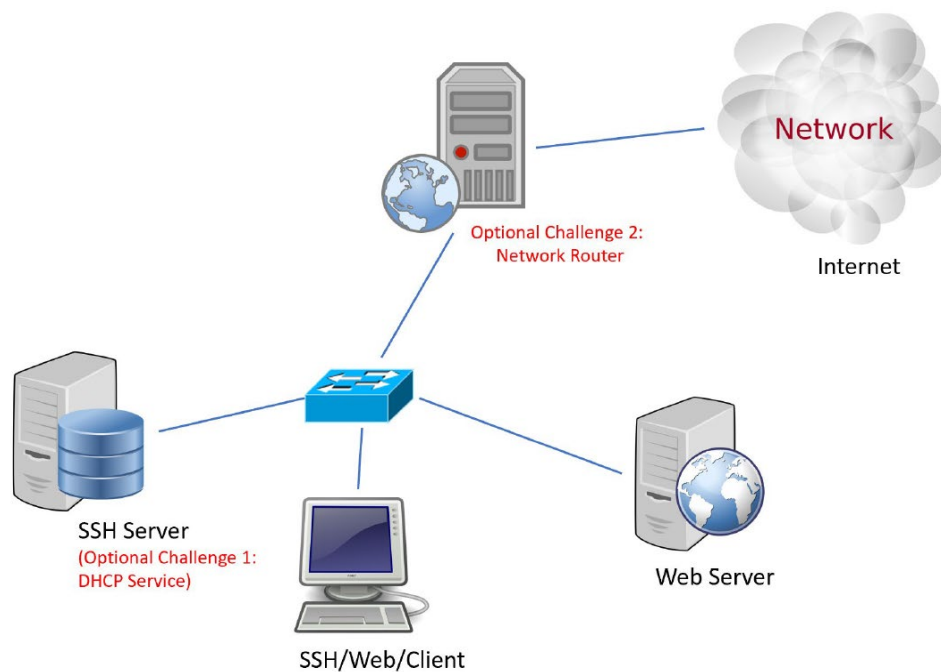


Figure 1.

Refer to the diagram (see Figure 1.) above. Your group of maximum 5 members is to set-up the network depicted in the diagram above.

Note that you are to propose your own IP addressing scheme. The following is suggested:

Server	# of Network Interfaces	IP Address Assignment
SSH Server	Only one (internal network)	Statically assigned
Client computer (SSH and Web)	Only one (internal network)	Statically assigned OR

		Optional challenge 1: Setup a DHCP server on either the SSH or Apache Web server and obtain an IP address dynamically / automatically from the DHCP service
Apache Web Server	Only one (internal network)	Statically assigned
Optional challenge 2: Set up a fourth Linux server that functions as a network router. Once this network router is successfully set-up, the SSH Server, Apache Web Server and Client computer machines should be able to access the Internet via the network router.		

Base test cases

- All three machines can successfully “ping” each other.
- Can successfully login to the SSH server from the Client computer Terminal.
- Can successfully access the default web page of the Apache Web Server from the Client computer browser.

Optional challenge 1:

- The client computer is automatically assigned an IP address from the DHCP service running on either the SSH or Apache Web Server.

Optional challenge 2:

- Show that all the other three machines can reach the Internet via the Network router linux server [note: you should demonstrate that the servers cannot reach the Internet when the network routing service is disabled or misconfigured]

Optional challenge 3:

SSH - Implement key-based authentication

Optional challenge 4:

Web - Implement Virtual Sites

Optional challenge 5:

DHCP client - Implement RAID 5 with 3 disks

Optional challenge 6:

DHCP - Implement IP reservation

Optional challenge 7:

Router - Implement NFTables to allow only web access and block ping to the Internet

You shall submit a group report (**Use the Template Provided**) documenting the steps to install, set-up, configure and tests the network connectivity and network services. As a guideline, the report should contain the following sections:

- Declaration of non-plagiarism
- Introduction
- Installation (include screen captures)
- Setup and/or Configuration (include screen captures)

- Testing (include screen captures)
- Conclusion
- References

You can assume that the target audience of your report is already familiar with the installation of the Linux operating system and hence do not need to include the installation steps.

Please note that you will be assessed individually hence do indicate the work or tasks that you've done in the group.

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Individual Contribution and Assessment

As this project is **assessed individually**, each group member must clearly indicate their personal contributions in the report. To do this:

- Include a **Contribution Table** (after the Declaration of Non-Plagiarism) listing each member's name and the specific sections or tasks they completed.
- Label each report section with the author's name (e.g., "*Section 3.2 written by: Ali*").
- Each member will be asked at least one question during the presentation about their declared work. Inability to answer may result in a mark reduction for Section 2.
- A **Peer Assessment form** will be distributed after submission. Scores from your groupmates may be used to adjust your individual mark.
- Optional challenges completed must also be attributed to a named individual in the Contribution Table to be eligible for bonus credit.

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2. Group Presentation (20%)

Each group will be allocated 20–25 minutes to present and conduct a live demonstration (with slides and demonstration of the above test cases and any relevant OS/Vbox configuration) Q&A will be approximately 5 - 10 minutes.

Please note that you will be assessed individually hence do indicate the work or tasks that you've done in the group.

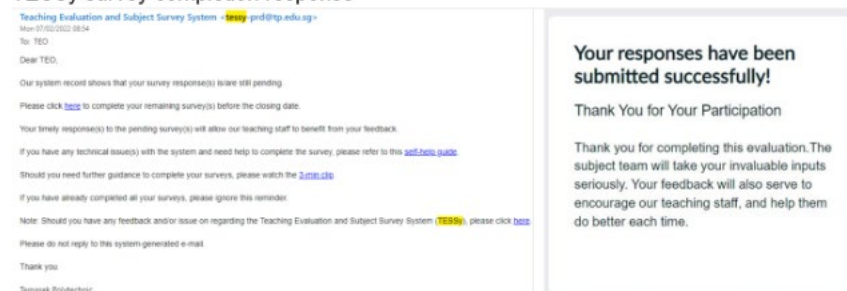
Deliverables

Week 16 (due 3rd August 2026, 8am)

- Submit group presentation slides / group report
- Prepare for project presentation according to your group allocated time slot.

Individual **TESSy Subject Survey** must be completed with screenshots of completion.

TESSy survey completion response



Or

<https://tp.bluera.com/tp/>



Late submissions

Late submissions will attract penalties:

- Late and < 1 day – 10% deduction from absolute marks given for the part of the work e.g., if the assignment was worth 100 marks, you were given 75 marks for the work, after the penalty you are left with 65 marks i.e., $75 - 100 \times 10\%$
- Late ≥ 1 and < 2 days – 20% deduction from absolute marks
- Late ≥ 2 days – no marks will be awarded

Feedback and questions

You are encouraged to obtain feedback from your tutors and ask questions related to the requirements of the Project.

Conduct

Be warned that students caught **plagiarising; including AI-generated work e.g., ChatGPT.** will have serious consequences. Please cite any sources used in your research.

===== END OF PROJECT SPECIFICATION =====